Food Delivery Analytics

Overview

An end-to-end data analytics project examining **15,000 food delivery orders** across **36 variables** to uncover operational inefficiencies, customer behaviour patterns, and revenue opportunities.

The analysis was completed entirely in **Google Sheets** using formulas, lookup tables, dashboards, and pivot tables—demonstrating how meaningful business insights can be generated without programming tools.

Business Questions

- What factors have the greatest impact on delivery performance?
- Which customer segments generate the most revenue?
- How do traffic and weather affect operational efficiency?
- Are loyalty and premium programmes driving meaningful behavioural change?
- Which areas should the business prioritise to improve customer satisfaction and profitability?

Project Structure

```
food-delivery-analytics/
│
├── README.md
├── INSIGHTS.md
├── .gi
│
├── data/
│   └── food_delivery_analytics_cleaned.csv
│
└── analysis/
   └── food_delivery_analysis_v2.xlsx
```

Dataset Overview

Property	Detail
Dataset	Food Delivery Orders
Records	15,000
Variables	36
City Tiers	Metro, Urban, Sub-urban
Time Period	12 Months
Revenue Analysed	$1.79M
Tooling	Google Sheets

Key Variables

Customer

- Age
- Loyalty Score
- Loyalty Tier
- Premium Customer Flag

Financial

- Order Value
- Delivery Fee
- Discount Amount
- Tip Amount
- Final Amount Paid

Delivery

- Delivery Time
- Delivery Distance
- Preparation Time
- Estimated Delivery Time

Operations

- Traffic Severity
- Weather Severity
- Delivery Efficiency Score
- Partner Experience

Flags

- Cancellation
- Delay
- Refund
- Promo Code Usage

Key Findings

- Generated **$1.79M** revenue from **15,000 orders**
- Average order value: **$113.95**
- Cancellation rate: **13.35%**
- Tier 3 cities generated **50.2% of total revenue**
- Premium customers spent **12.2% more per order**
- High traffic increased delivery times by approximately **13 minutes**
- Severe weather increased delivery times by approximately **11 minutes**
- Loyalty tier showed minimal impact on customer behaviour
- Late-night orders experienced the highest cancellation rates

See **INSIGHTS.md** for the complete analysis and business recommendations.

Tools & Techniques

Tools

- Google Sheets

Functions

- COUNTIF / COUNTIFS
- SUMIF / SUMIFS
- AVERAGEIF / AVERAGEIFS
- IFERROR
- XLOOKUP
- IFS
- Pivot Tables

Analytical Methods

- KPI Analysis
- Customer Segmentation
- Cohort Comparison
- Correlation Analysis
- Cross-tabulation
- Trend Analysis

Methodology

1. Reviewed and validated all source variables.
2. Created lookup tables for customer, traffic, weather, and experience bands.
3. Built KPI dashboards using formula-driven calculations.
4. Performed segmentation analysis across customer and operational groups.
5. Created pivot tables to identify trends and relationships.
6. Conducted correlation analysis to identify key delivery drivers.
7. Converted findings into prioritised business recommendations.

Deliverables

- Interactive Excel Workbook
- KPI Dashboard
- Pivot Table Analysis
- Business Insights Report
- Strategic Recommendations